



UNIT 04

PROGRAMMING SCRATCH

Student Learning Outcomes:

After the completion of this unit students will be able to:

- What is Computer Program
- Program & Programming
- Programming Languages
- Importance of Programming language
- Basic Programming concept
- Differentiate between Algorithm and Program
- Getting Starting with Scratch programming
- Setup Scratch Software
- The Scratch Editor
- Programming Sprite
- Fundamentals Of Scratch Constructs
- Design your first Scratch Program
- Running a program
- Debugging a program



Introduction to the Unit

In this chapter, we will learn about the fundamentals of computer programming. We will analyze and apply basic constructs (e.g, sequence, selection, repetition, variables, inputs/events) by creating short programs using a visual programming tool.

4.1 Program Concept

4.1.1 Program

A computer program is like a set of instructions that tells a computer what to do. It is a series of steps or commands written in a language that the computer can



Fig 4.1

understand. These instructions guide the computer to perform tasks efficiently.

4.1.2 Programming

Computer programming is the process of creating and writing a computer program. Programming is like giving instructions to a computer so it can do things for you. It is a bit like writing a recipe for a cake: you tell the computer what steps to follow, and it does them to achieve the required outcome.

4.1.3 Programming Language

Programs are written in programming languages, which provide the facility to write a program in text-based editor or in visual mode. Programming languages are classified into different types which are as under.



Figure 4.2

Do You Know?

ENIAC was the first electronic computer that could run programs, but programmable computers like we have today, the Manchester Baby, built in 1948, is regarded as the first one.

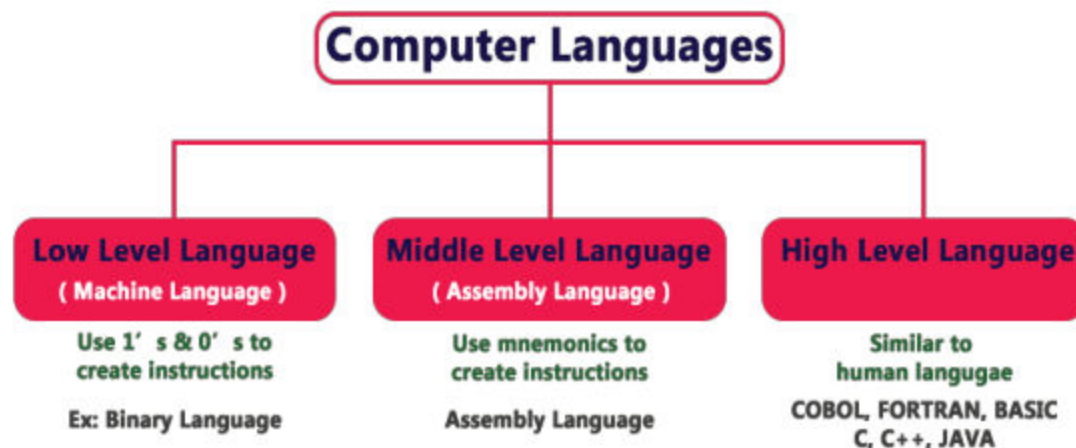


Figure 4.3

4.2 Importance and uses of Programming Language

In today's world, programming languages are important because they help to create useful things on computers like websites, games, phone apps, and more. Businesses and industries worldwide are changing and automating their operations. Many businesses need web designers, AI (Artificial



Figure 4.4



Figure 4.5

Intelligence) engineers, and mobile app developers for tasks like marketing, supply chain management, and customer care service. Computer programmers can work from anywhere for any business. Without programming languages, many of the things we use daily would not exist.

4.3 Difference between Algorithm and Program

Algorithm	Program
- It is the conceptual idea of how to solve a problem. - It is like a recipe or step by step solution of a problem.	- It is a set of instructions written in a specific programming language. - It is the implementation of an algorithm.

- | | |
|--|--|
| <ul style="list-style-type: none"> • High level plan or strategy for solving a problem. • Describe in simple human language or with diagram. | <ul style="list-style-type: none"> • It uses syntax and features of programming language. • It is actual code that translates algorithm into instructions a computer can understand and execute. |
|--|--|

4.3.1 Converting An Algorithm Into Program

Algorithm is created in the view of the problem under consideration. Therefore, we simply code the steps of algorithm in programming language like Scratch.

Steps of Algorithm	Steps in Scratch Program
Problem statement: <u>find the square of the given number.</u> Input A A=5 Multiply A with A Result 25	Step 1:  Step 2:  Step 3:  Step 4: 

You can try running this program in the “**Scratch editor**” by dragging these code blocks and clicking on the green flag to see how it works?

4.4 Introduction to Scratch Programming



Scratch is a block-based visual programming language and online community that provides an interface to create interactive stories, animations, and games. Instead of typing, users can use colorful blocks like puzzle pieces to build their creations. Scratch is free for anyone to use and share. When students create

Programming Scratch

programs, they learn important mathematical and computer concepts, which help them think creatively, solve problems, and develop logical reasoning. Scratch allows students to share their creations online, enabling them to show case their work and team up with others from around the world.



Figure 4.6

Note for Teachers

Teacher's Tip:

Teacher should explain the scratch online community to the students and how to join the scratch community.

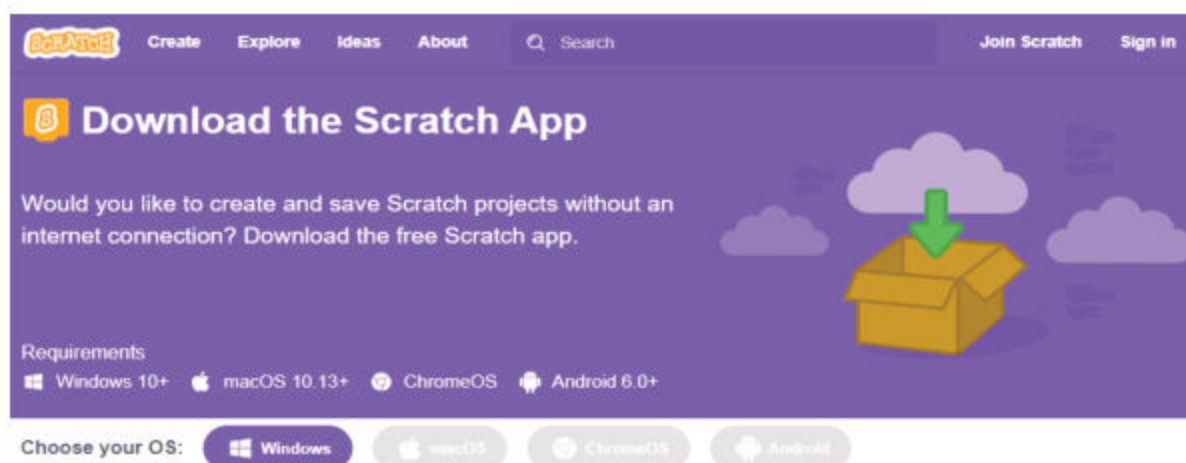


4.4.1 Downloading And Installing the Scratch Software

Step 1: Open Google Chrome or any other browser that you have in your PC and type the following URL

<https://scratch.mit.edu/download>

Step 2: Select **Direct Download** option as shown in the figure.



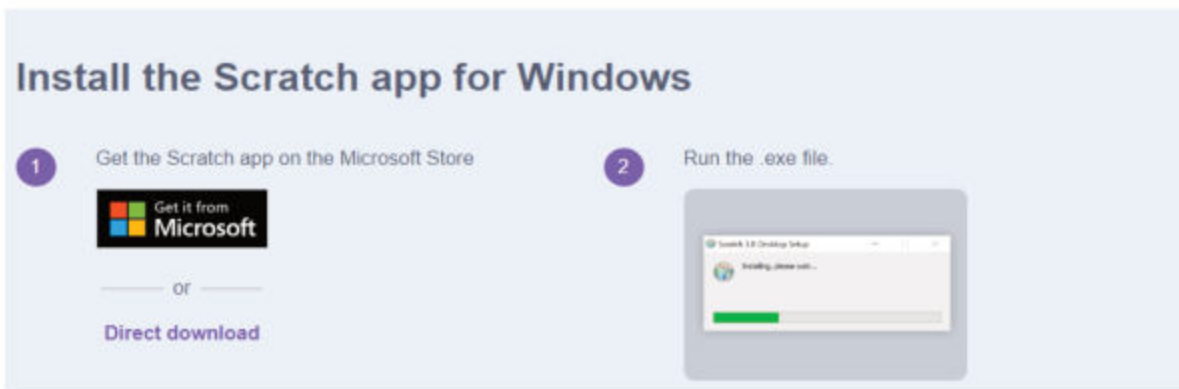


Figure 4.7

Step 3: After downloading the software, proceed with the installation steps.

Step 4: After successful installation process the Scratch icon appears on the desktop of your computer as shown in the figure. Click on the icon to start your Scratch software.



4.4.2 Using Scratch Online

You can also use Scratch online by accessing its editor through your web browser. Simply enter the following URL.

<https://scratch.mit.edu/projects/editor/>

Once you've opened the URL, Scratch will work just like the offline/downloaded version. You can also create an account on Scratch website to save your projects online.

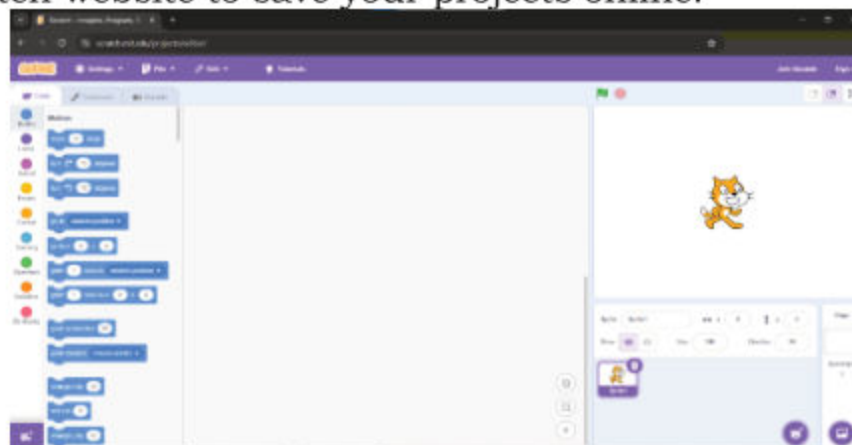


Figure 4.8

4.5 Working with Scratch



To start working with Scratch, double-click on the Scratch icon available on your computer desktop after installation. Once the software is opened you can see a Scratch editor screen.

4.5.1 Scratch Editor

The Scratch editor provides a platform where you can drag and drop snap code blocks to create games and animations. On the Scratch editor platform, we can find many options like the following.

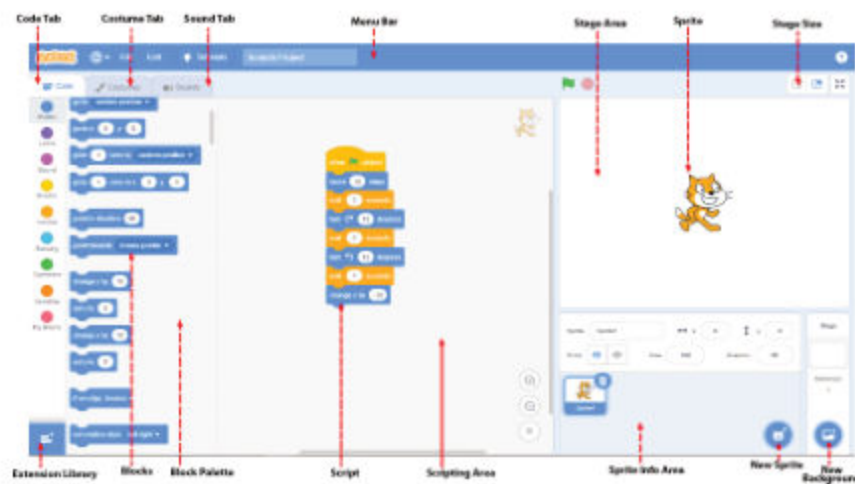


Figure 4.9 Scratch Editor



ACTIVITY

Teacher guide the students to open the Scratch software and explains all the options available in Scratch editor.

4.5.1.1 Costume Tab

It enables users to alter the appearance of a sprite to generate many effects, including animation. You can also choose a different costume by clicking the bottom-left button labeled “Choose a Costume”.

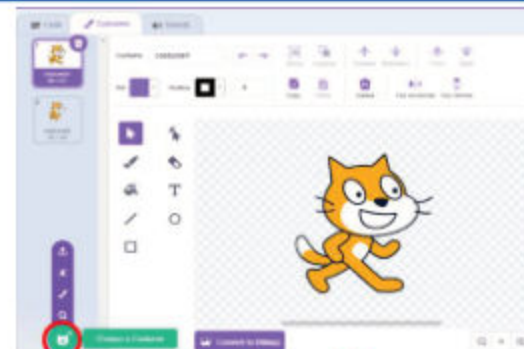


Figure 4.10

4.5.1.2 Sound Tab

This tab is used to record, import, or edit sounds. You can choose the sound from Scratch library, record your own sound, select a surprise sound or import a sound file from your computer.

Common Ribbon Menu Control



Upload: you can take a picture from your computer.

Surprise: Any random image will automatically select.

Paint: you can paint or drawn your creation.

Choose from library: you can select any image from available library.

These are the common menu items for Sprite, Sprite costume, Backdrop and Sound.

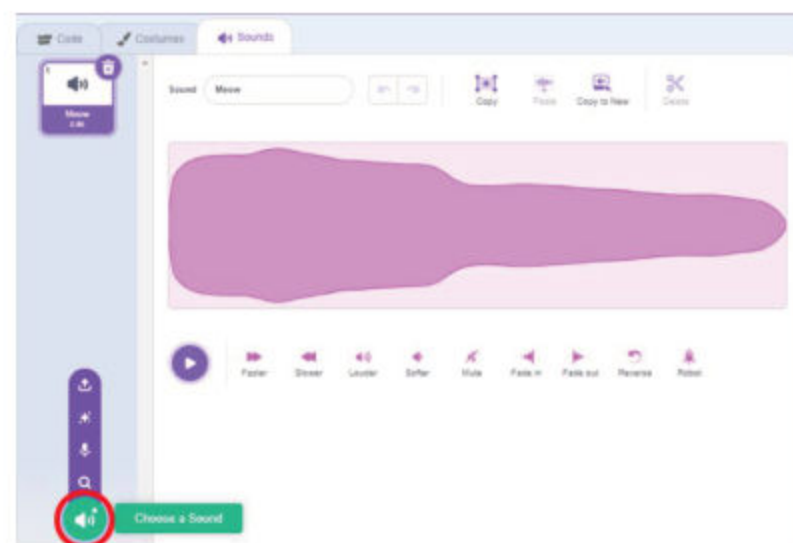


Figure 4.11

4.5.1.3 Code Tab

 Code Tab contains block Palette which have Control blocks used to control sprites. Programming a sprite always begins with a control block (orange or yellow category).

There are 9 block palettes available on the Code Tab.

1. **Motion:** When you click on the motion button, you'll find blue blocks to make your sprite move in different ways.
2. **Looks:** This purple block palette allows to change the color, size and costumes of your sprite.
3. **Sound:** This orchid color block palette allows you to add the sound in your program from the available sound library, record sound, or upload it from your computer.
4. **Events:** Event triggers specific code at a particular time or action. Such as clicking on green flag. These blocks are “starting blocks” means they must be placed at the beginning of each new code segment and found in yellow color.
5. **Control:** As the name control it used to control the scripts. It provides option to wait, use loops and conditional statements in your program.
6. **Sensing:** This block helps your program to detects things. They can find out where the mouse pointer is, how far apart sprites are, and if one sprite is touching another.
7. **Operators:** The operators block use to compare variables and values, do calculations with numbers, and work with strings(text).

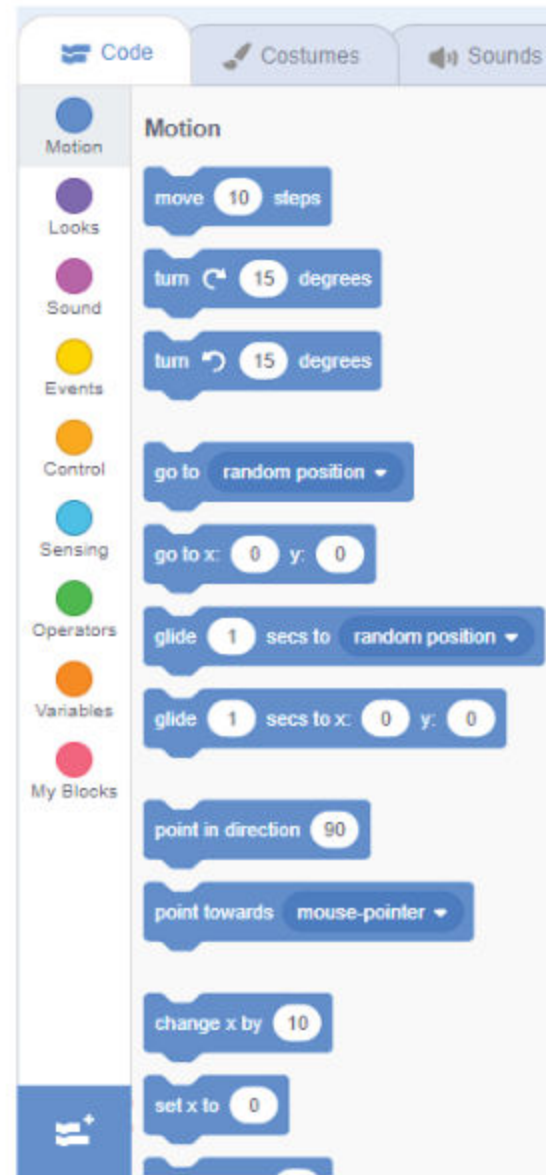


Figure 4.12 Code Blocks

8. **Variables:** The variable blocks are used to save the values that are used in the program.
9. **My Blocks:** This will allow to create new blocks for a sprite. You give the name and then define what the new block does using other Scratch blocks.

ACTIVITY

Teacher guide the students to open the Scratch software and explain each block one by one, providing examples for better understanding.

4.5.2 Elements Of Scratch

Scratch has four main parts: the **Stage**, **Sprites**, **Scripts**, and **Programming Palette**. We can think of them as the different roles and props in a play.

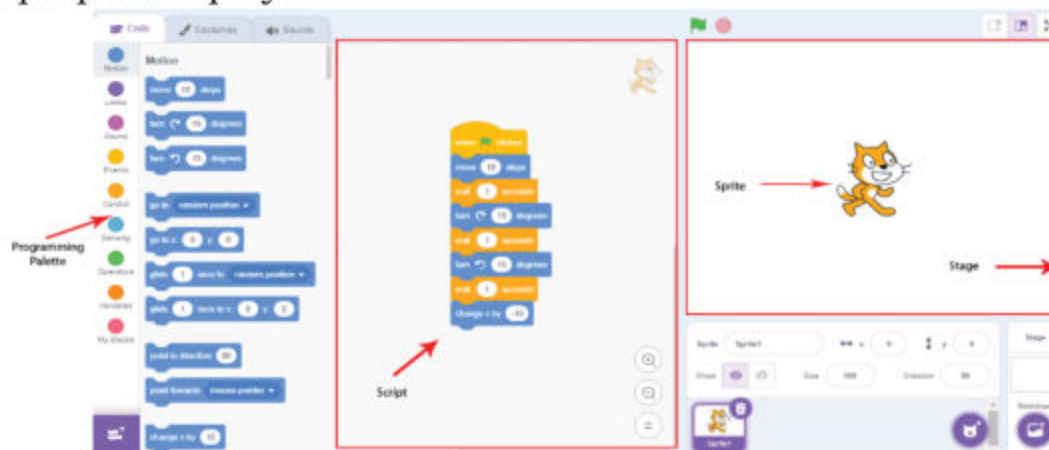


Figure 4.13

Sprite

They are the basic object in Scratch which are the actors or main characters of the project. Sprites are programmed to do something in Scratch. The default sprite in scratch is orange cat.

Stage

It is similar to the stage in a play. It is the main area where the action of your program takes place.

Script

It tells the actors what to say or do. Each sprite is programmed with a script.

Programming Palette

Elements used to program the sprite to do or say something. Sprites must be programmed to carry out every function you want them to perform.

4.6 Programming a Sprite

Programming a sprite means choosing a character or object, changing its appearance, position, background, and foreground colors and telling it what to do by giving it instructions in the form of block code which looks like a puzzle that you simply drag and drop from Block Palette.

4.6.1 Choose Sprite

The default sprite is an orange cat. You can choose or draw a different sprite from the **Choose a Sprite** option located at the bottom of **Sprite Info Area**. You can select any sprite from the library or draw your own or upload from your computer.

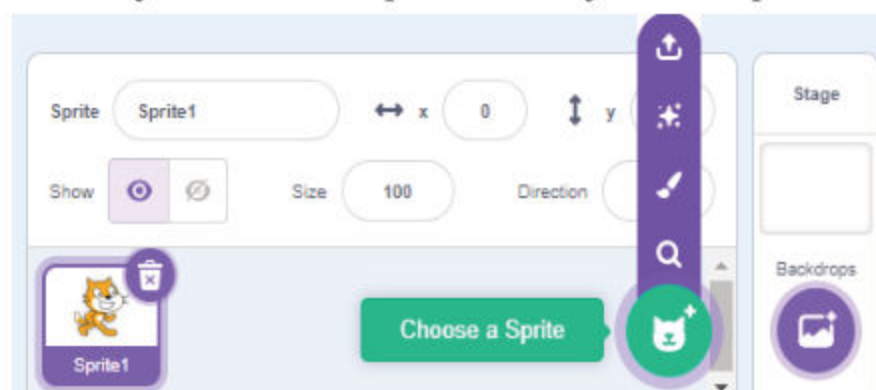


Figure 4.14

ACTIVITY

In the Scratch editor, replace the default sprite with a new one that you like.

4.6.2 Choose a Backdrop

A backdrop is like a background of your stage. The **Choose a Backdrop** option is available at the bottom right corner next to the **Choose Sprite** button. You can choose backdrop from library, you can paint on your own using Scratch paint editor, surprise option will select random backdrop from the library and upload option allows to upload an image from your computer.

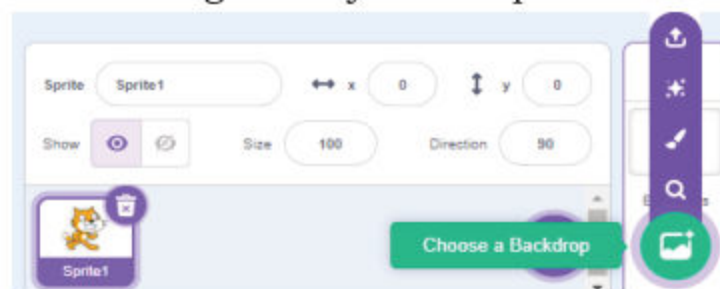


Figure 4.15

ACTIVITY

Upload a new background from your computer in the Scratch stage area.

4.6.3 Stage Area

The Stage Area is the main area where all the fun activities happens. It is where characters move perform actions, and follow the instructions you give them. The stage acts as the background of your project. Coordinates in Scratch are based on the grid image below.



Figure 4.16

Note

Coordinates are like addresses that tell you exactly where something is located. They're used to pinpoint a specific position on a map, grid, or other surface.

X and Y coordinates are like the address of a point on a graph. The X coordinate tells you how far to go horizontally (sideways), and the Y coordinate tells you how far to go vertically (up or down).

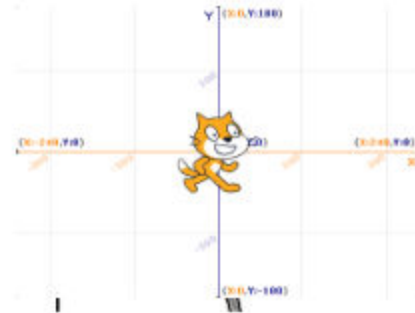


Figure 4.17

Stage Adjustment:

1. Small stage - decrease the stage windows and increase the script area.
2. Regular size - increase the stage windows and decrease the script area.
3. Full screen mode - switch the stage window to full-screen.

4.7 Fundamentals of Scratch Constructs

Programming constructs are simple building blocks that help you control how your program works.

There are five main types of programming constructs in scratch.

1. SEQUENCE

In programming, a sequence means a list of steps that tell the computer what to do, one after the other. In Scratch, if you have two characters, each can have its own list of steps to follow and do something.

2. EVENTS

An event is something the program knows about and reacts to, like pressing a key, clicking a mouse, or

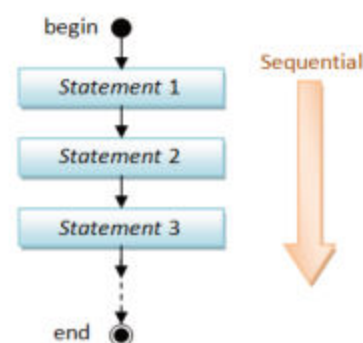


Figure 4.18

tapping a phone screen. In Scratch, events are shown in yellow color.

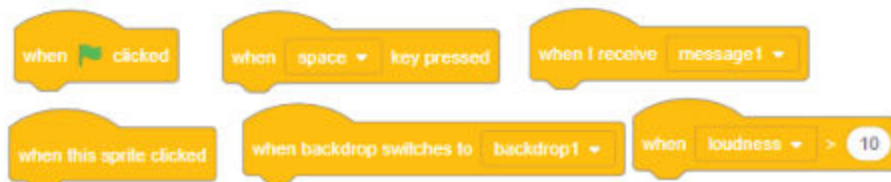


Figure 4.20

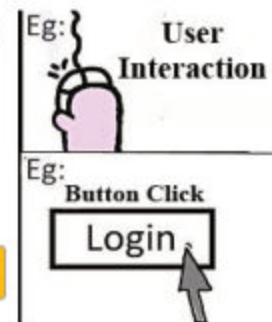


Figure 4.19

3. LOOPS

Loops are like a magic trick in coding that make action repeat again and again without needing to write the code multiple times. In the control section of block code, you'll find three types of loops, each with its own special job.



repeat (number n) Block

The "repeat" block is a type of control block. It lets you repeat a set of commands 'n' times. The number 'n' can be any positive whole number.



repeat until Block

The "repeat until" block is a type of iteration block. It runs a set of commands until a certain condition remains true. The condition you put inside "until" decides when the loop stops.



forever Block

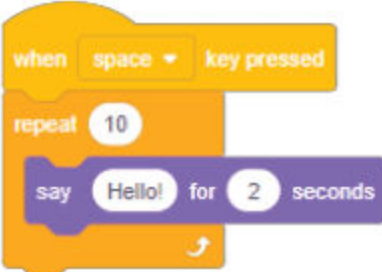
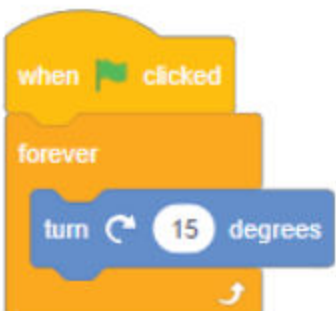
The "forever" block is a type of iteration block. It runs a set of commands continuously without stopping. There's no stopping



Programming Scratch

condition in the forever block. It is used when you want something to keep happening forever, like a never-ending point.

Examples of Loops

Type	Code Block	Program Script	Description
Repeat x Number of times			When the space bar on the keyboard is pressed, the Sprite will say "Hello" every 2 seconds and repeat it 10 times.
Repeat Until			When you click on Sprite it will say "Hassan is Cool" and repeat it every 2 second until the space bar on keyboard is pressed.
Forever			When Green Flag is clicked Sprite will turn 15 degree clockwise forever.

ACTIVITY

How many times does these loops repeat the block inside?



4. SELECTION/ CONDITION

A condition is like a rule in a program that tells it what to do next. It is similar to a yes-or-no question. If the answer is "yes" (true), the program does one action. If the answer is "no" (false), it does something else. Conditional statements are like everyday choices. Computers use them to decide what to do next based on whether something is true or false. You'll see them in control blocks of code tab.

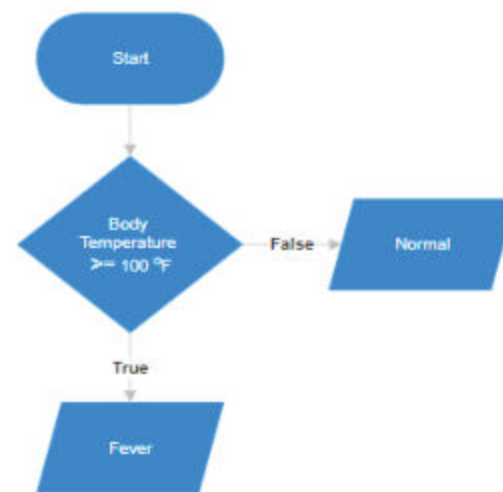


Figure 4.21

If then Block

It is a control block. This block works on the condition's value. This block decides what to do depending on a condition. If the condition is true, it runs some code. If not, nothing happens. For example, as shown in **Fig 4.22**, if the user types "a", then the sprite must glide for 1 second.

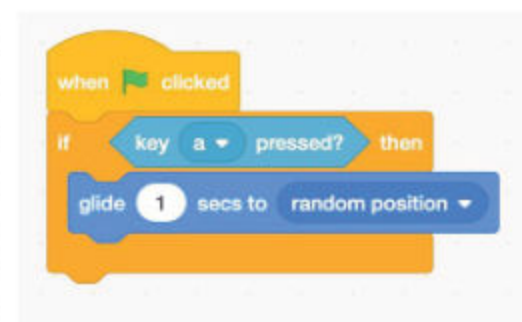


Figure 4.22

If then else Block

It is a control block. This block is an extension of the if-then value. If condition is true, then a set of lines specified in “If” body gets executed. And a condition is false, then set of lines specified in “else” part is executed. For example, if the user types “a”, then the sprite must glide for 1 second or else must say “hello”.

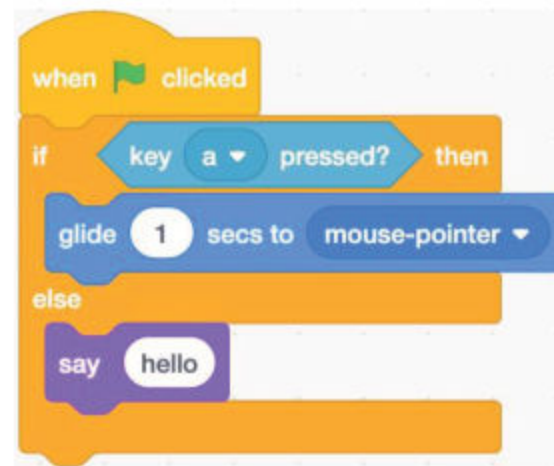


Figure 4.23

5. VARIABLES

Variables are like containers that hold values. Imagine a box with a number inside it. A variable is a named computer's memory location where we can store data. It can only hold one piece of information at a time. Variables store data of different types (like name, number/digits etc.). Once the variable is created, we can use it in different parts of a program.

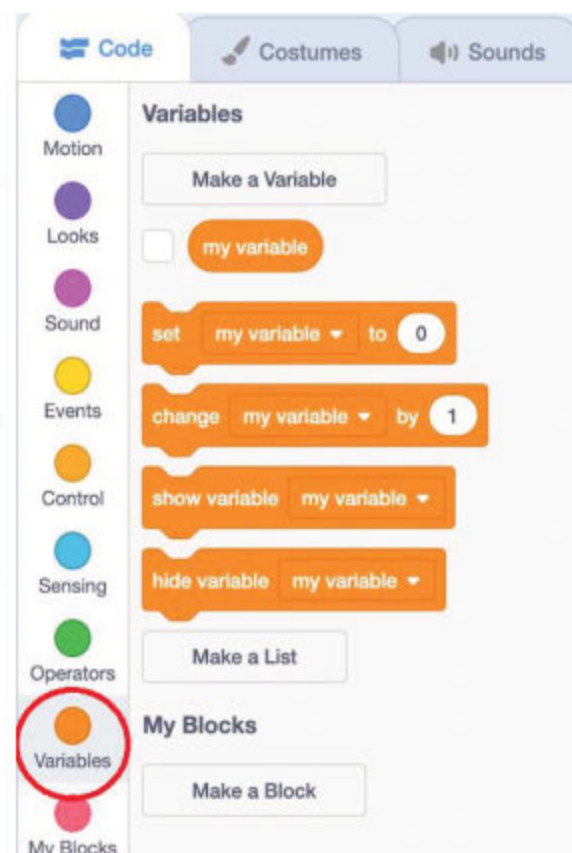


Figure 4.24

4.8 Create and Run First Program in Scratch

Now it is time to create the first simple program in Scratch by using different code blocks. In this program, our default sprite will interact by asking question and showing a greeting message. Let's begin with following steps.

4.8.1 Creating First Program

Open Scratch software's editor

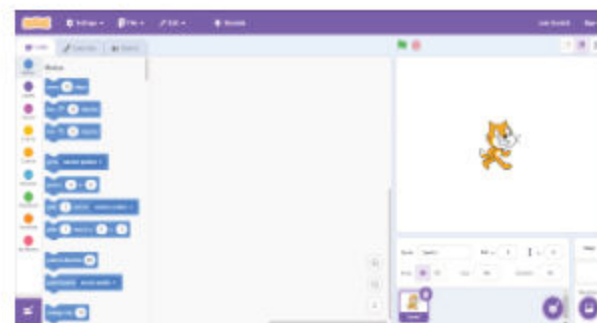


Figure 4.25

1. Select your default sprite or you can change the sprite.
2. Go to **Events** block palette and



drag  “**when Green flag Clicked**” block in scripting area.

3. Go to **Sensing** block palette and drag “**ask what's your name? and wait**”  block in scripting area.

Here you will see that both blocks fit into one another like puzzle or Lego pieces.



Figure 4.26

4. Go to the **Looks** blocks palette (purple) color and drag **say Hello! for 2 seconds** and snap it right below of the previous block.

4.8.2 Saving First Program

To save your program, go to the File menu and click on Save to Your Computer.

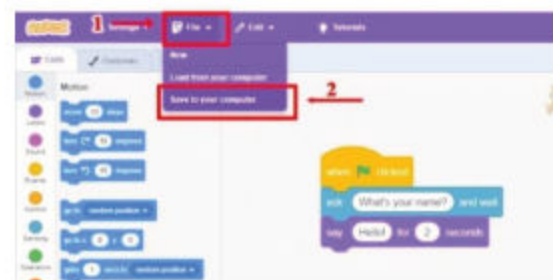


Figure 4.27

4.8.3 Running First Program

To run the first program, follow the steps:

1. Click on the green flag button present on the top side of **stage Area**.

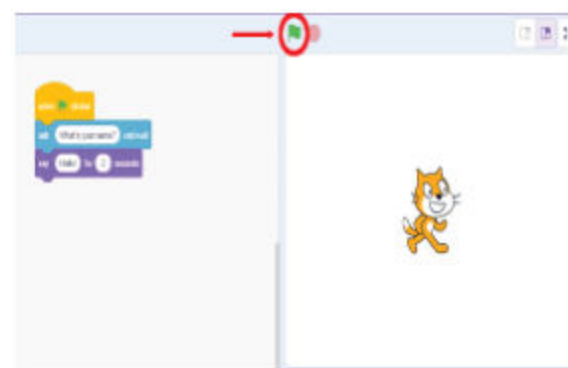


Figure 4.28

- The Sprite will ask for the name by prompting "What's your name?".



Figure 4.29

- Type your name in purple highlighted text box and press Enter from your keyboard.

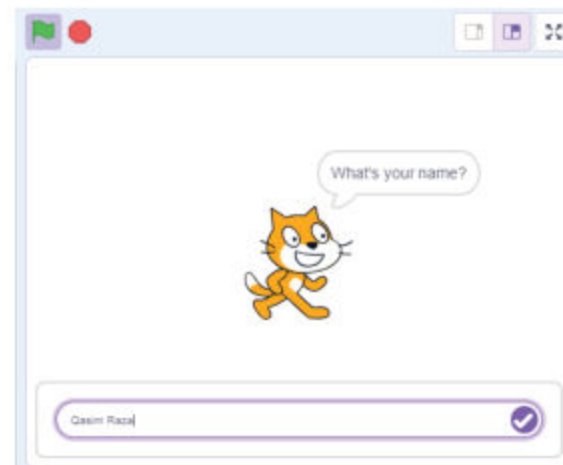


Figure 4.30

- The cat will say hello and after two seconds the bubble disappears.

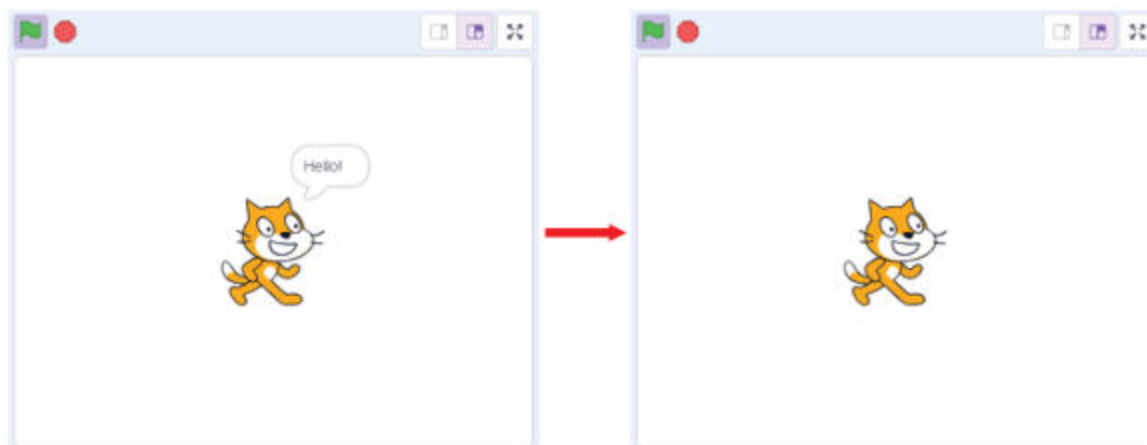


Figure 4.31

This is our first program which says the following when the green flag is clicked the cat (sprite) will ask for the name and then wait after we input our name the cat will say hello for two second. This is how code works these code blocks are executed one after another.

4.8.4 Make First Program Smarter

Now the sprite will do something a little bit smarter this time, so let's go to **Sensing** code block palette which is in light blue color.

1. Open your first program from your computer. Go to file menu and click "Load from your Computer".



Figure 4.32

2. Your first saved scratch program is now open into your editor.

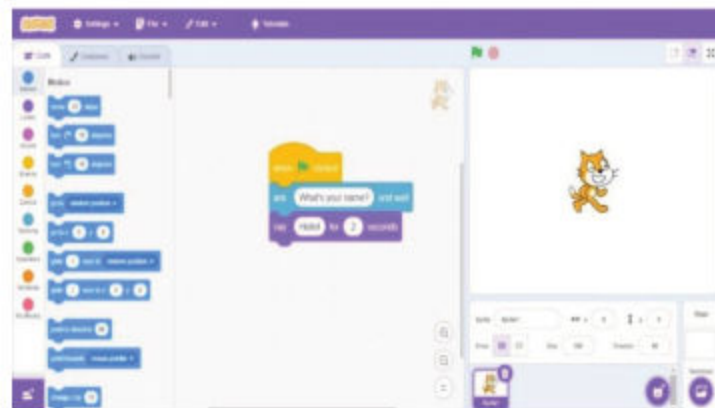


Figure 4.33

3. Drag **answer** block and snap it into “**say Hello! for 2 seconds**” block where “Hello” is showing.



Figure 4.34

4. Now cat/sprite instead of saying hello it is going to say the answer whatever you type in the text box in previous step **fig 4.8.3 3**.

5. Run the program, typing the name and now sprite will say the name.

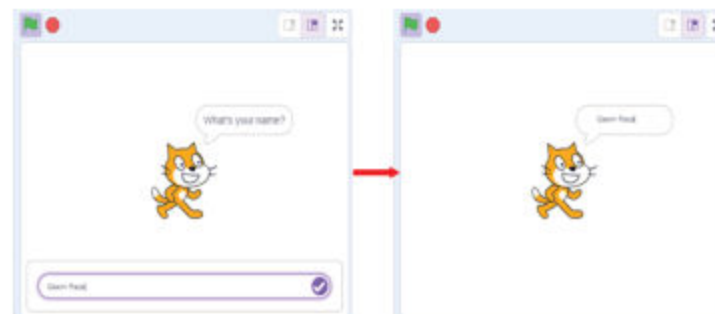


Figure 4.35

By doing such code our sprite will echo back whatever we typed. Let's make the sprite do a proper greeting by saying “Hello Qasim Raza” instead of saying “Qasim Raza”. For this we use **Operators** code block palette.

6. Go to **Operators** block palette (green) and drag the **join** block code into scripting area.

7. Take out the answer

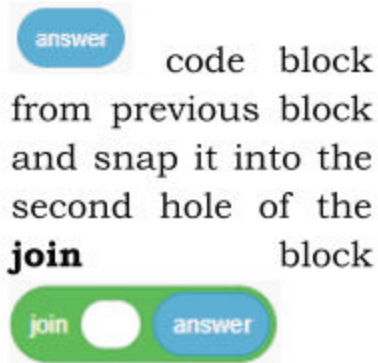


Figure 4.36

8. In first hole of **join** block type some greetings word like hello and snap this join block to the previous “say” block.

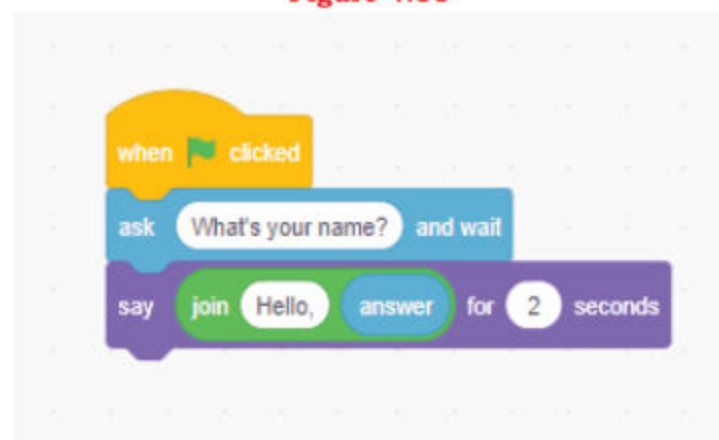


Figure 4.37

Congratulations, at this point your first code is smarter than previous one. Now run your program and see the results.

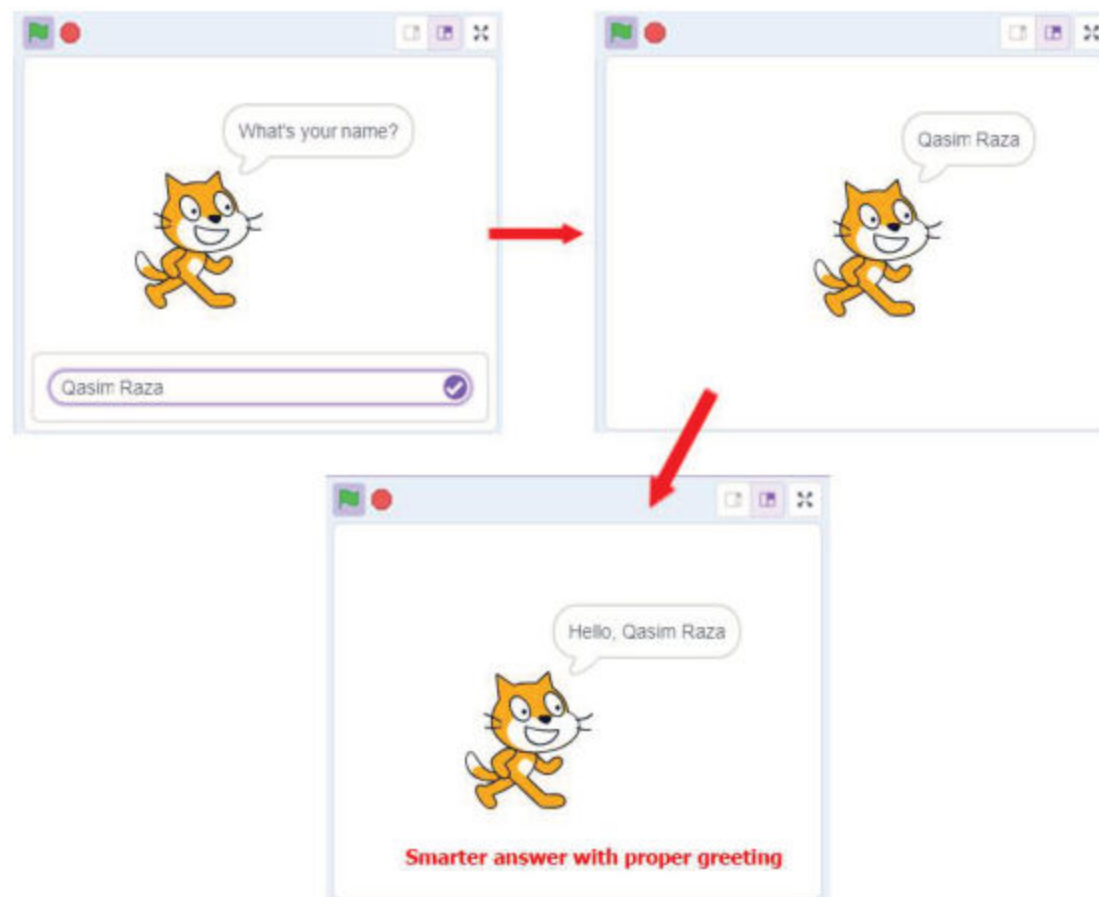


Figure 4.38

4.9 Debug A Program

Bugs are errors or mistakes in writing a program. Debugging a program means finding and fixing mistakes or problems in it, so that program will run smoothly. It is like solving puzzles to make sure the program works correctly. Following is the debugging cycle through which we can fix the bugs.

- Run the program.
- Understand the problem that causes program to run abnormally.
- Examine the code to identify where things might be going wrong.
- Fix the issue in the code.
- Run the program again to make sure the problem is fixed.

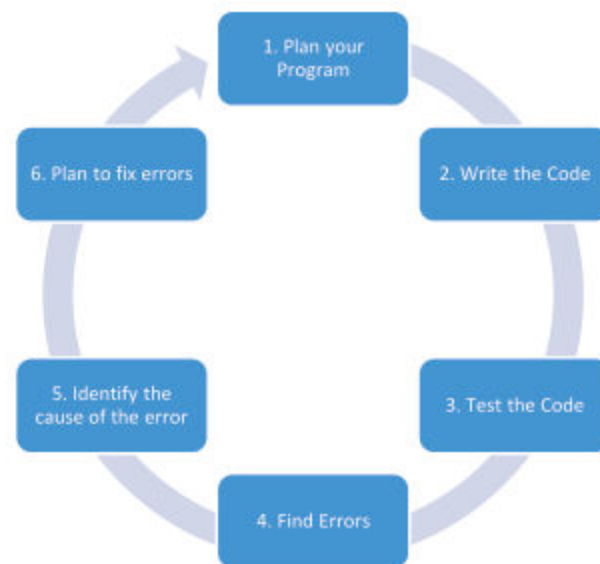


Figure 4.39

4.10 Creating And Running Another Program

We are creating another program in which Sprite will move from left to right and right to left.

- Open Scratch Software.
- Select the favorite sprite.

Select your desired sprite by clicking on the "**Choose Sprite**" button. A sprite selection window will appear. Select your favorite sprite, and it will be added to the sprite area. Once your sprite is added, a small cross icon (x) will appear next to the default sprite, allowing you to delete it.

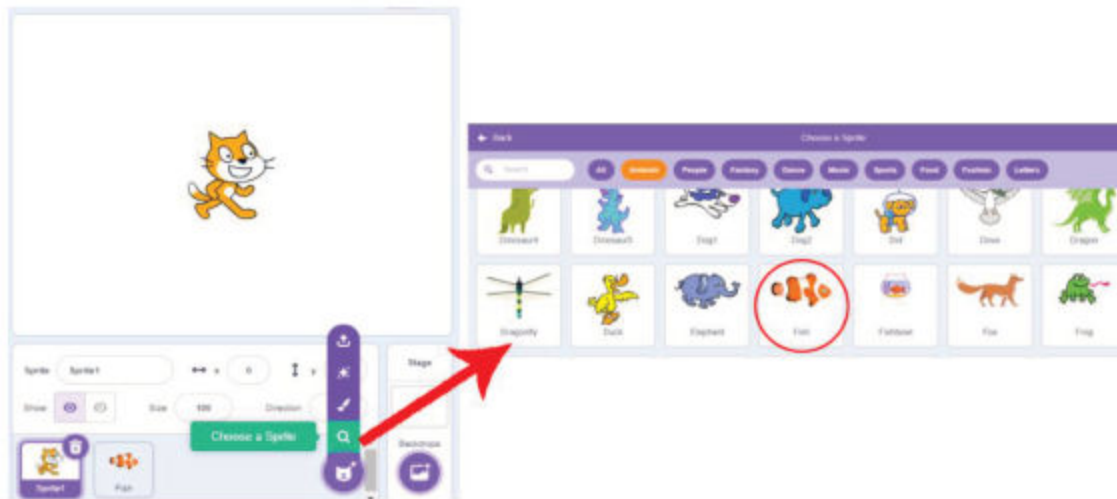


Figure 4.40

If you want to change the color of your sprite, you can go to the costume tab and change the color, use any available costume of your sprite, or design your own.

Costume
Tab

See Fig 4.14

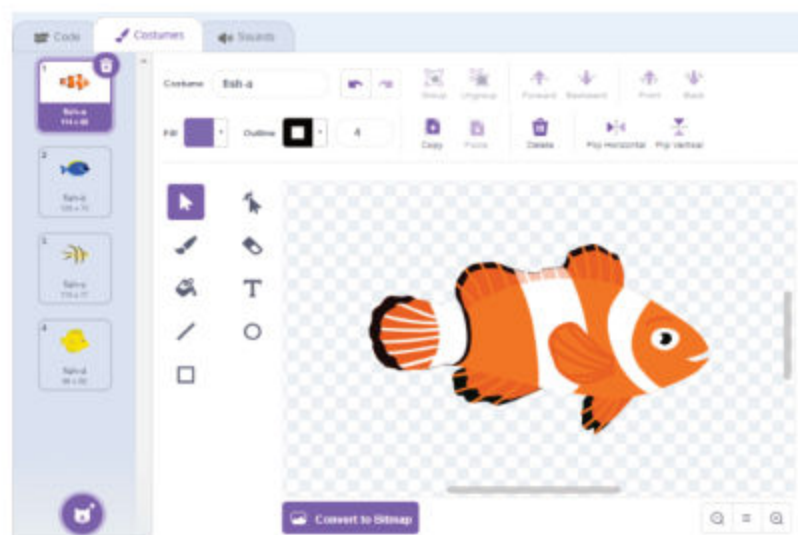


Figure 4.41

- Select any suitable backdrop (if you want)

Backdrop

See Fig 4.15

Select the new backdrop from choose a backdrop option.

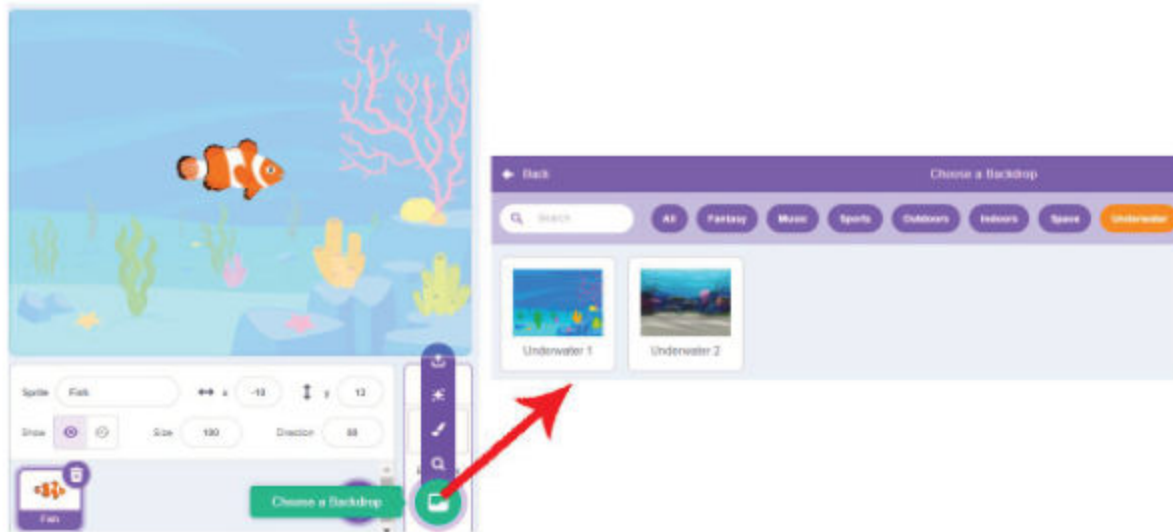


Figure 4.42

All the basic steps are done now it is time to program some code.

Click on the sprite to write its script so the sprite should move. Follow these below steps.

- go to Events block palette and drag  “when Green flag clicked” code block in scripting area.
- go to Motion block palette drag  “Move 10 Steps” code block to make your sprite move 10 steps.

By writing above code, sprite will only move 10 steps and then stop. To make the sprite move continuously use a forever loop.

- drag the **forever loop** block from **Control** block palette and snap “**move 10 steps**” block into the **forever loop** block.



Here, you can see that because of the “**forever**” loop block, the sprite will continuously move and may move outside the stage area. So, you have to use the “**if on edge, bounce**” block

from **Motion** block palette to bounce back the sprite in the stage area.

When your sprite is back on the stage, it appears vertically and horizontally flipped, as shown in the figure.

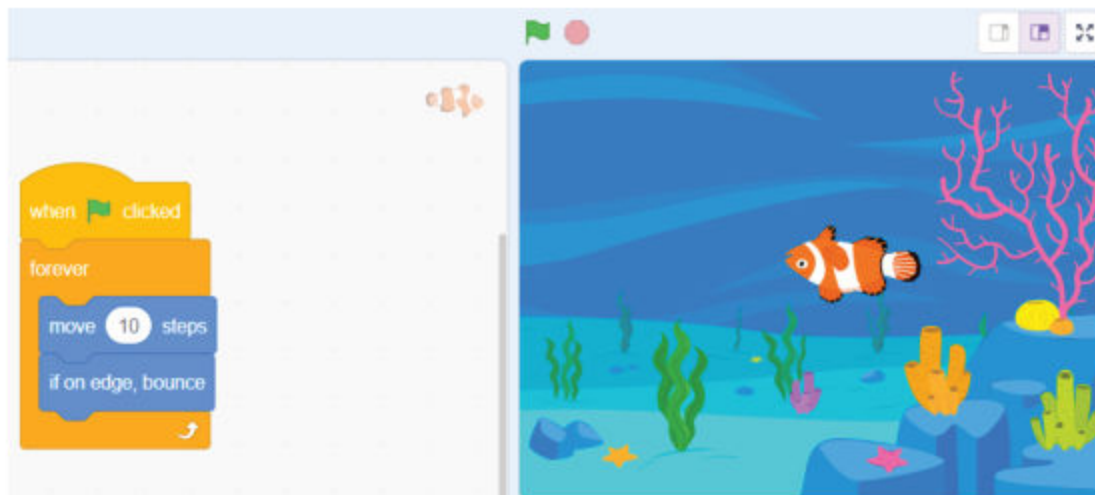
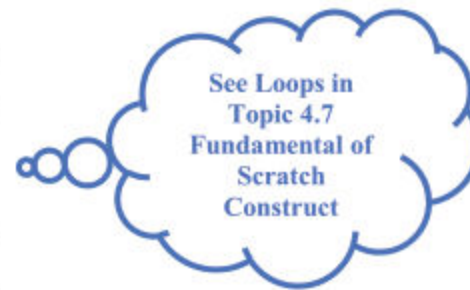


Figure 4.43

To set the sprite orientation after it bounces back from the edge.

- Drag the “**set rotation style, left right**” block from **Motion** block palette.

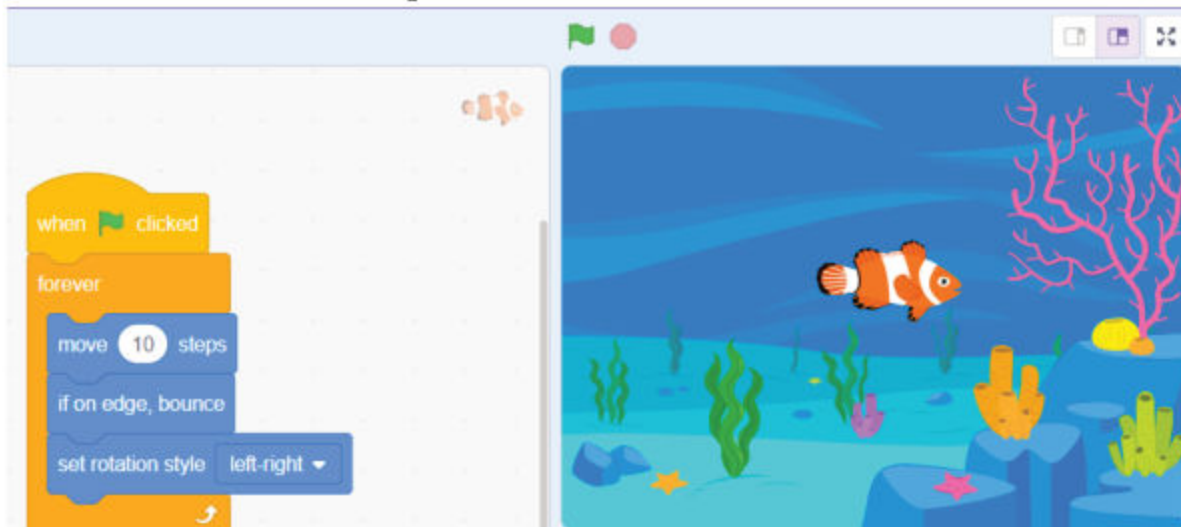


Figure 4.44

You will see the sprite move from right to left and back again. You can control the speed of sprite by adding

- The “**wait**” block from **Control** palette and set the desired time in second instead of 1.

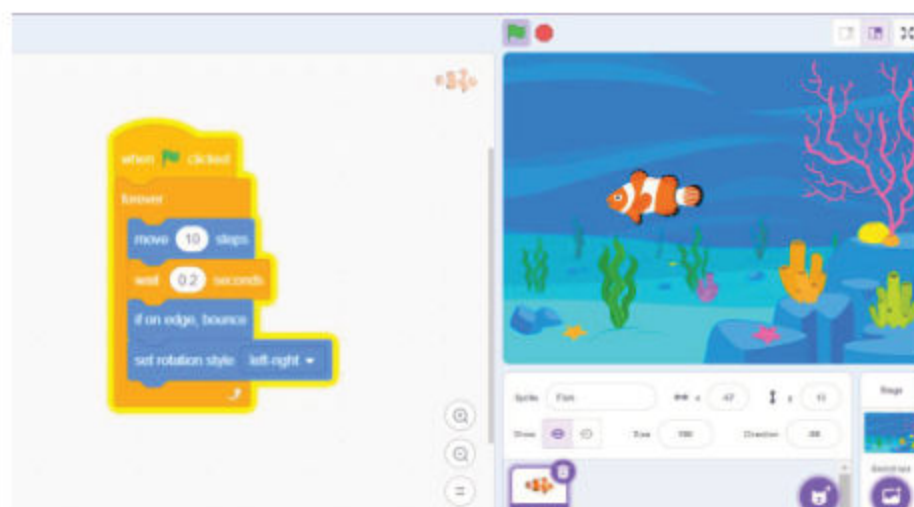


Figure 4.45



We can make this program more interactive by adding additional blocks based on the situation. For example, when the sprite bounces back from the edge, its costume changes. Additionally, before bouncing off the edge, the sprite can say "Bye," and after bouncing back, it can say "Hello."

- Use the **Sensing** block and **Control** block to detect the edges of the stage and apply condition if-else block.

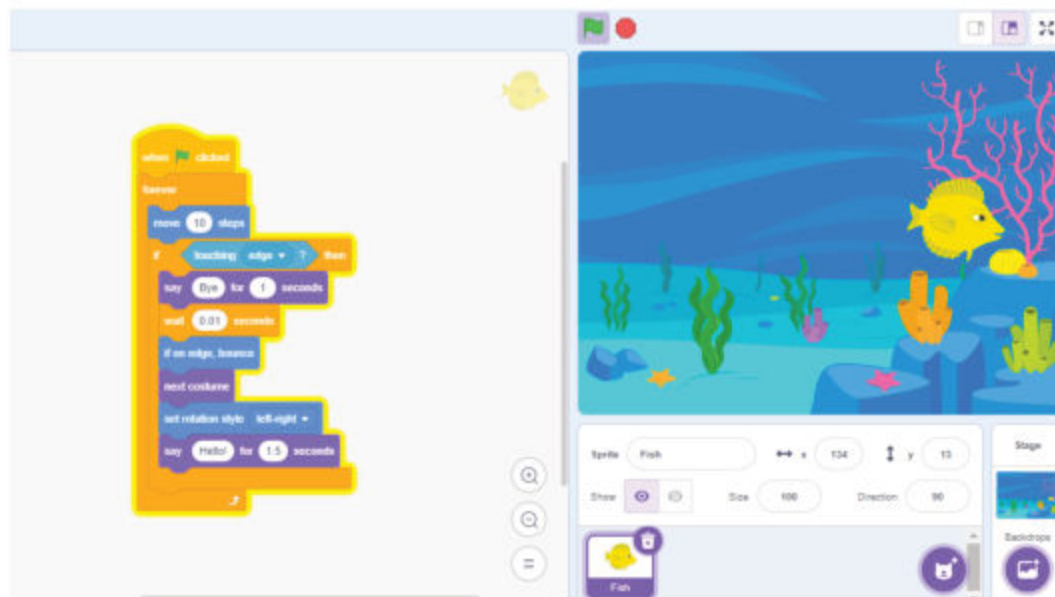


Figure 4.46



Update your code as shown in the above picture and click on the green button to run the program.

Congratulation! You have created your first animated Scratch program.

Summary

- ✓ Scratch programming is an engaging way for kids and beginners to create their own stories, games, and animations.
- ✓ The main purpose of Scratch is to help young learners learn basic coding concepts without getting into the difficulties of textual coding. Here are some benefits of coding with Scratch.
 - Enhances analytical and problem-solving skills
 - Makes it easier to learn coding
 - Makes learning fun and interesting
 - Explains programming logic visually
- ✓ An **Event** is like a signal that makes something happen. Like pressing button to start the game.
- ✓ A **Sequence** is a list of actions performed one by one, in a specific order. Computers follow these actions exactly as they're written.
- ✓ A **Loop** repeats a set of instructions until a specific stopping condition is met. It is a way to do something over and over again without having to manually do it each time.
- ✓ A **Variable** is like a container where we keep some information/data, which we call its value.
- ✓ An **Error** results in incorrect execution or behavior of a program.
- ✓ **Debugging** is the process of finding errors or bugs in a program and fixing them so that the program can execute smoothly.
- ✓ A **conditional statement** asks a question to figure out which path to take next.



Important Terms to Remember

Online Community: An online community is like a big group of people who talk and share stuff on the Internet. It is like a virtual version of a neighborhood or club where people with similar interests or goals come together to chat and help each other out.

MIDI: Musical Instrument Digital Interface

URL: Uniform Resource Locator

Program: Set of instructions

Repetition/Iteration: Act of doing one thing again and again

Add-on: An add-on is a small program or tool that adds extra features to existing software, enhancing its functionality.

Dragging: Pull (something) along forcefully

Constructs: to make or form by combining or arranging parts or elements

Extension Library: An extension is an add-on that adds extra features and functions to a program or software. It improves the original capabilities, allowing users to do more with the software.

Exercise

1. Encircle the correct answers.

- i) VPL is the:

a. Graphical Programming Language	b. Set of instruction
c. Visual Programming Language	d. A Background
- ii) Scripts in programming are the:

a. List of sprites	b. List of background
c. A set of instructions blocks connected with one another used for programming.	
d. None of the above	
- iii) The types of code blocks in Scratch are the:

a. 3	b. 5	c. 7	d. 9
------	------	------	------
- iv) The blocks which we use to start the program:

a. Control block	b. Event Block
c. Motion Block	d. Looks Blocks
- v) The actual working space where you can drag and drop Code blocks to be executed is:

a. Block palette	b. Backdrop
c. Script Area	d. Stage

2. Fill in the blanks with appropriate words given below.

Repeat, Wait, if – else, Use the move block, Use the if block

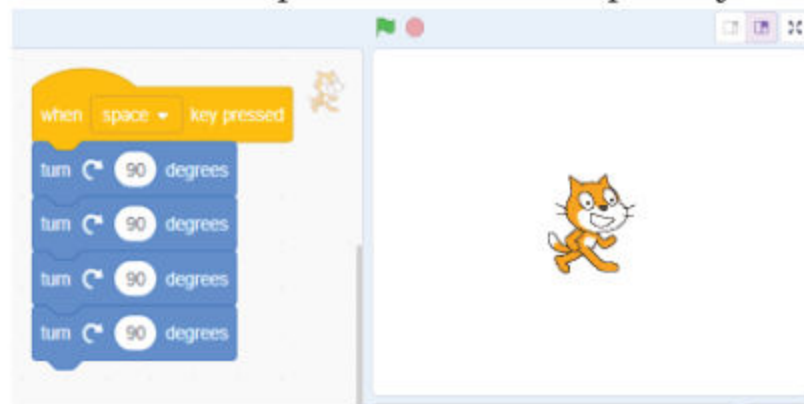
- i. The block used to create a loop in Scratch is _____.
- ii. The construct used for creating a wait time in Scratch is _____.
- iii. We check if a certain condition is true in Scratch using _____.
- iv. The construct used for making decisions in Scratch is _____.
- v. To move a sprite in Scratch, we _____.

3. Provide descriptive answers to the following questions.

- i. What is programming?
- ii. Differentiate between Algorithm and Program.
- iii. Explain Sprite, Script and Stage.
- iv. Define 9 available options in block palettes.
- v. What is control block used for?
- vi. In what situations are conditional statements used?
- vii. How do we change the backdrop in Scratch?

4. ACTIVITY

- i. Get the Scratch Software on your lab computers by following the instructions in the software download section and install it.
- ii. Write a code that makes the sprite moves 10 steps only and make this sprite move in the direction for 10 times, what code block should be added to the code?

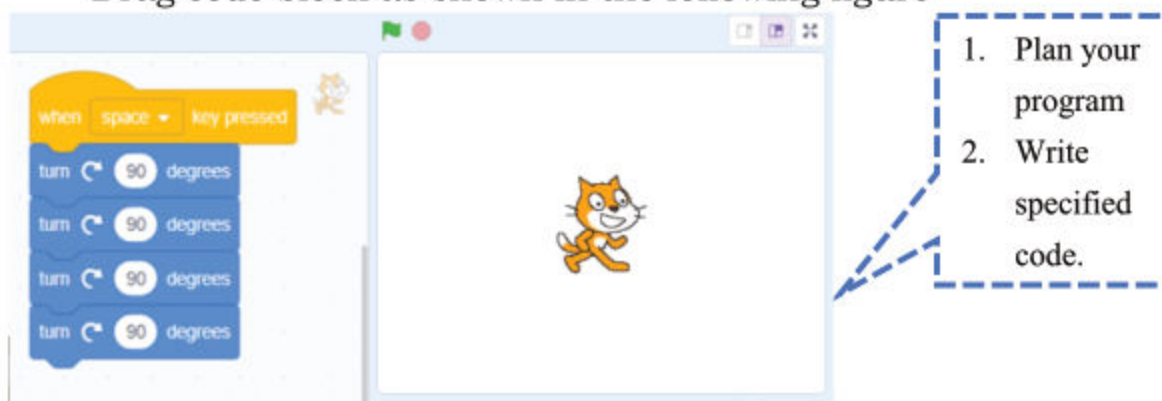


- iii. Design a game which using cursor key to move the sprite on the stage from left to right and right to left.
- iv. Teacher should guide students how to fixing the bug in the code by apply debugging process

APPLYING DEBUGGING PROCESS

Open your scratch editor

Drag code block as shown in the following figure



Run your code and see the results.
Nothing will happen

The block codes run at a same time one after another in micro seconds.

Use **Control** block palette and

drag **wait**  code block.

The block codes run at a same time one after another in micro seconds.

Use **Control** block palette and

drag **wait**  code block.

1. Plan your program
2. Write specified code.

3. Test the code

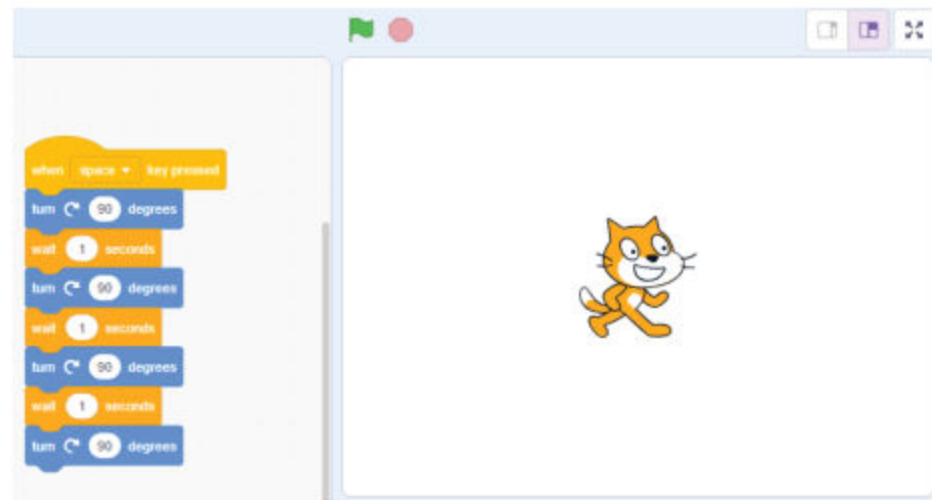
5. Identify the cause of error.

6. Plan to fix errors.

7. Identify the cause of error.

8. Plan to fix errors.

Programming Scratch



Note for Teachers

- Teachers should motivate students to explore various resources and continue learning independently.
- Divide the class into group of 2 or 3 students.
- Open all the code block option from each block palette and discuss with groups about the working of each and every block by placing the blocks in the Scratch script area.
- Make a sample program in which two Sprites talk with each other. Try making few jokes.

USEFUL LINKS

- <https://scratch.mit.edu>
- <https://scratch.mit.edu/download>
- <https://scratch.mit.edu/projects/editor/>
- <https://scratch.mit.edu/ideas>
- <https://scratch.mit.edu/explore/projects/tutorials/>